Task 3: Reflection on AI Paradoxes

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As a student studying Agentic AI for Business and FinTech, I have gradually realized that the development of artificial intelligence involves several interesting paradoxes. AI systems are becoming increasingly capable of performing complex tasks, yet greater intelligence and automation do not necessarily lead to greater reliability, security, or trust.

The first paradox is that greater AI capability does not always guarantee greater reliability. In Homework 2, I developed a CV verification agent that used DeepSeek, LangChain, and MCP tools to compare CV information against LinkedIn and Facebook profiles. The agent achieved 100% accuracy on the public test set and successfully detected several deliberately modified CVs. However, I eventually created an adversarial PDF that visually displayed a false graduation year while providing the correct year through an invisible text layer. The agent classified this CV as valid because the PDF parser extracted the correct year rather than the false information visible to a human reader. This experience showed me that even an agent capable of accurate reasoning can produce incorrect conclusions when its input representation is misleading.

The second paradox is that automation can increase efficiency while creating a greater need for human oversight. In financial services, AI agents can support KYC, compliance monitoring, and risk assessment by processing large volumes of information more efficiently than manual workflows. However, when automated decisions are based on incomplete or manipulated evidence, errors can spread quickly and affect many users. Therefore, I believe human oversight should not disappear as AI becomes more autonomous. Instead, it should shift toward reviewing unusual cases, evaluating evidence quality, and ensuring accountability.

The third paradox is that making AI systems more useful can also increase their vulnerability. Agentic AI becomes powerful when it can access external tools, retrieve information, and act independently. Yet every additional tool, data source, or processing component introduces potential weaknesses. My homework demonstrated that the vulnerability was not necessarily in the language model itself, but in the PDF extraction process preceding verification. This suggests that securing an AI agent requires examining the entire workflow rather than focusing only on the model.

Overall, this course has changed how I evaluate AI systems. I previously focused mainly on whether an AI model could produce accurate results. Now, I believe it is equally important to understand how those results are generated, what assumptions the system makes, and how it behaves when encountering unexpected or adversarial inputs. As AI becomes more widely adopted in financial services, balancing

efficiency, reliability, and human accountability will be essential for building trustworthy applications.